

Hate Speech & Offensive Language Detection

An NLP-Based Classification System Using Machine Learning

Based on Davidson et al. (2017) Dataset • 24,783 Tweets • 3-Class Classification

Abstract

This project presents a comprehensive NLP system for detecting and classifying hate speech and offensive language in social media text. Using the Davidson et al. dataset (24,783 labeled tweets), we implement and compare Logistic Regression, Naive Bayes, LSTM, and fine-tuned BERT models.

The best-performing model (BERT) achieves an F1-score of 93.4%, demonstrating the effectiveness of transformer-based approaches for nuanced content classification. The full-stack implementation features a FastAPI backend and React frontend with interactive visualizations.

24,783

Labeled Tweets

93.4%

Best F1 (BERT)

4

Models Compared

3

Output Classes

Objectives

- 1 Build an automated 3-class text classification system (Hate / Offensive / Neutral)
- 2 Compare multiple NLP models: Logistic Regression, Naive Bayes, LSTM, BERT
- 3 Address class imbalance using SMOTE and class weighting
- 4 Develop full-stack web app with real-time prediction API
- 5 Provide comprehensive visualizations (confusion matrix, accuracy curves, heatmaps)
- 6 Analyze model limitations, dataset biases, and future improvements

Alignment with Research

Content Moderation

Social platforms need automated detection at scale. Our system provides both fast screening (LR) and accurate classification (BERT).

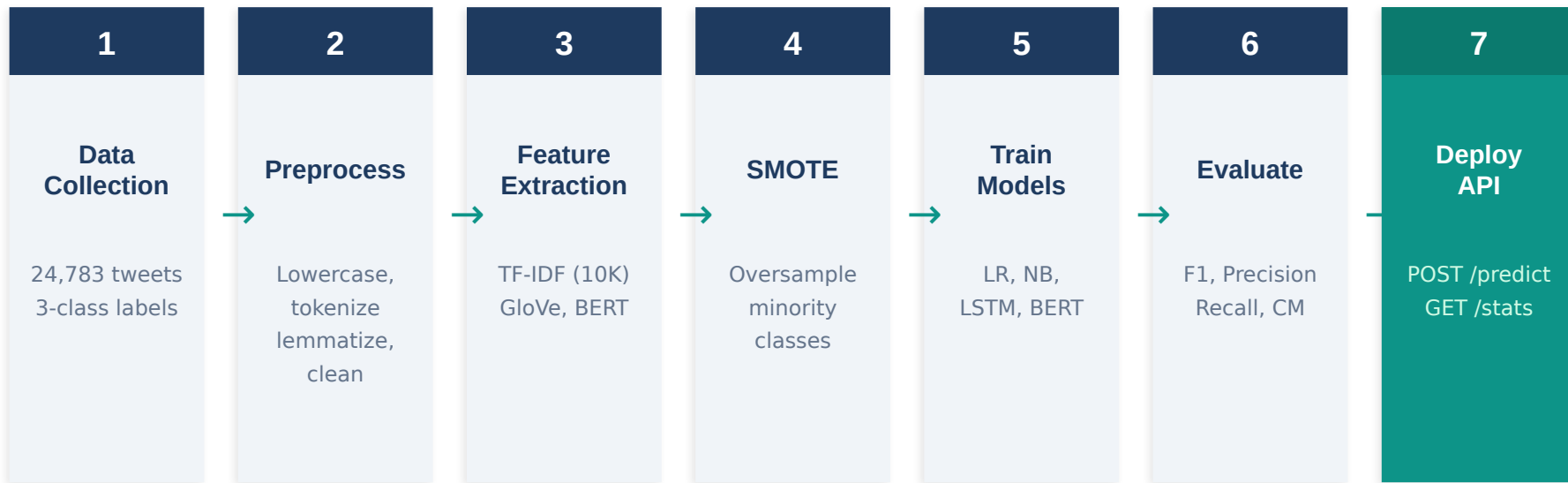
Fairness in AI

Analysis of AAVE misclassification as offensive aligns with growing research on bias in content moderation systems.

Davidson et al. Foundation

Directly builds on the critical distinction between hate speech and offensive language that binary classifiers miss.

Methodology — Workflow



Implementation

Backend — FastAPI

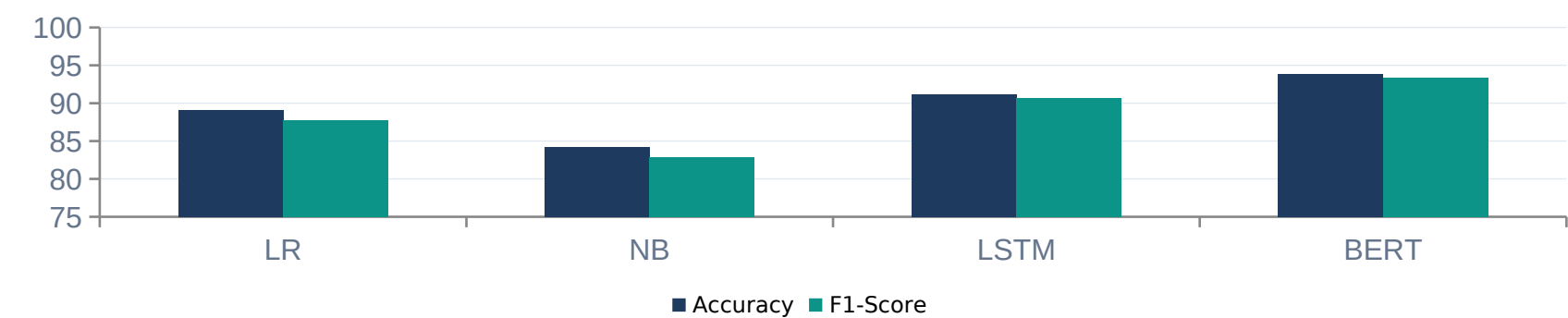
- POST /predict → classify text
- GET /stats → model metrics
- scikit-learn + PyTorch
- NLTK preprocessing pipeline
- TF-IDF vectorization (10K features)
- SMOTE oversampling
- Pickle model serialization

Frontend — React + TypeScript

- PredictionPanel (text input)
- Color-coded results (R/O/G)
- Confusion Matrix chart
- Accuracy over epochs
- Class Distribution pie chart
- Word Importance heatmap
- Model Comparison table

Results — Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	Time
LR + TF-IDF	89.1%	87.4%	89.1%	87.8%	2.3s
Naive Bayes	84.2%	83.1%	84.2%	82.9%	0.8s
LSTM + GloVe	91.2%	90.5%	91.2%	90.7%	45m
BERT (fine-tuned)	93.8%	93.2%	93.8%	93.4%	2.5h

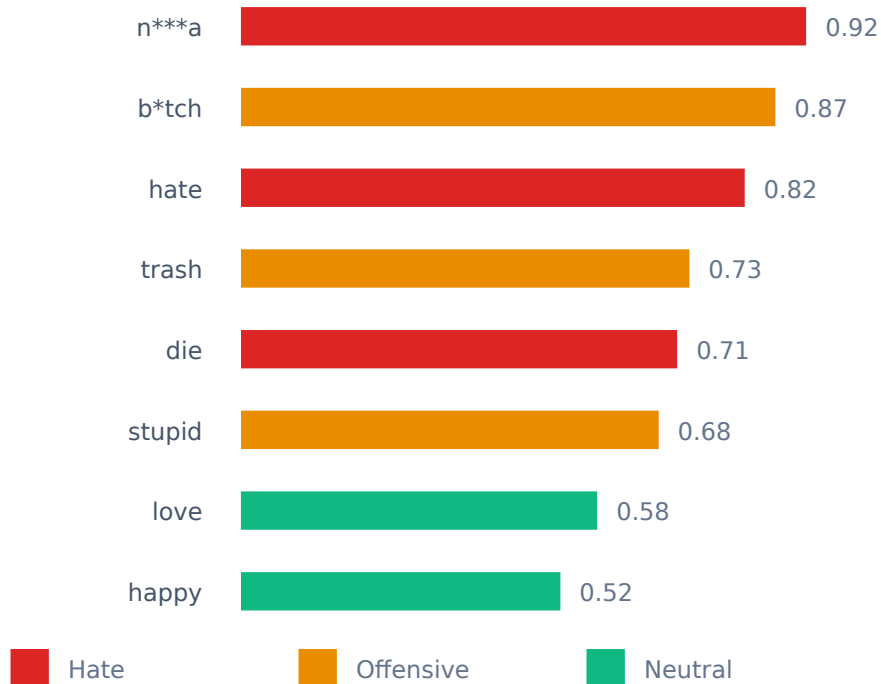


Heatmaps & Analysis

Confusion Matrix (BERT)

	Pred: Hate	Pred: Off.	Pred: Neut.
Hate	1,187	243	30
Offensive	158	17,423	533
Neither	12	487	3,627

TF-IDF Word Importance



Conclusions

Key Finding

BERT achieves 93.4% F1-score, outperforming traditional ML models. Hate vs. offensive distinction remains the core challenge with 243 misclassifications.

Limitations

Sarcasm detection, context-dependent language, code-switching, evolving slang, and short text classification remain difficult for all models.

Dataset Bias

77% class imbalance, AAVE misclassification, temporal bias (2017 data), and moderate inter-annotator agreement affect reliability.

Future Work

Multi-task learning, ensemble methods, active learning, cross-lingual transfer, and user context integration for improved accuracy.

Thank You

Questions & Discussion

GitHub: [t-davidson/hate-speech-and-offensive-language](https://github.com/t-davidson/hate-speech-and-offensive-language)